

## Problem

If  $g(t) = 4e^{-2t} \cos 4t$ , find  $G(s)$ .

## Step-by-step solution

## Step 1 of 4

Consider the following signal:

$$g(t) = 4e^{-2t} \cos(4t)$$

The Laplace transform of  $g(t)$  is

$$L\{g(t)\} = L\{4e^{-2t} \cos 4t\}$$

$$G(s) = L\{4e^{-2t} \cos 4t\}$$

$$G(s) = 4L[e^{-2t} \cos 4t] \dots\dots (1)$$

## Step 2 of 4

First, determine the Laplace transform of  $x(t) = \cos 4t$

$$L[x(t)] = \int_0^{\infty} x(t) e^{-st} dt$$

$$X(s) = \int_0^{\infty} \cos(4t) e^{-st} dt$$

It is known that,

$$\cos(\omega t) = \frac{e^{j\omega t} + e^{-j\omega t}}{2}$$

Therefore,

Therefore,

$$\begin{aligned} X(s) &= \int_0^{\infty} \left( \frac{e^{j4t} + e^{-j4t}}{2} \right) e^{-st} dt \\ &= \frac{1}{2} \left[ \int_0^{\infty} e^{j4t} e^{-st} dt + \int_0^{\infty} e^{-j4t} e^{-st} dt \right] \\ &= \frac{1}{2} \left[ \int_0^{\infty} e^{-(s-j4)t} dt + \int_0^{\infty} e^{-(s+j4)t} dt \right] \\ &= \frac{1}{2} \left[ \left[ \frac{e^{-(s-j4)t}}{-(s-j4)} \right]_0^{\infty} + \left[ \frac{e^{-(s+j4)t}}{-(s+j4)} \right]_0^{\infty} \right] \end{aligned}$$

Further simplification as follows,

$$\begin{aligned} X(s) &= \frac{1}{2} \left[ \left[ \frac{e^{-(s-j4)t}}{-(s-j4)} - \frac{e^{-(s-j4)(0)}}{-(s-j4)} \right] + \left[ \frac{e^{-(s+j4)t}}{-(s+j4)} - \frac{e^{-(s+j4)(0)}}{-(s+j4)} \right] \right] \\ &= \frac{1}{2} \left[ \left[ \frac{0}{-(s-j4)} - \frac{1}{-(s-j4)} \right] + \left[ \frac{0}{-(s+j4)} - \frac{1}{-(s+j4)} \right] \right] \\ &= \frac{1}{2} \left[ \frac{1}{(s-j4)} + \frac{1}{(s+j4)} \right] \\ &= \frac{1}{2} \left[ \frac{s+j4+s-j4}{(s-j4)(s+j4)} \right] \\ &= \frac{1}{2} \left[ \frac{2s}{s^2+16} \right] \\ &= \frac{s}{s^2+16} \end{aligned}$$

### Step 3 of 4

Refer to Table 15.1 for properties of the Laplace transform in the text book.

The frequency shifting property of Laplace transform is,

$$L[e^{-at} f(t)] = F(s+a)$$

and

$$L[e^{-at} \cos bt] = \frac{s+a}{(s+a)^2 + b^2}$$

Therefore, the Laplace transform of  $e^{-2t} \cos 4t$  is,

$$\begin{aligned} L[e^{-2t} \cos 4t] &= \frac{s+2}{(s+2)^2 + 4^2} \\ &= \frac{s+2}{(s+2)^2 + 16} \end{aligned}$$

Recall equation (1),

$$G(s) = 4L[e^{-2t} \cos 4t]$$

Substitute  $\frac{s+2}{(s+2)^2+16}$  for  $L[e^{-2t} \cos 4t]$  in the equation.

$$\begin{aligned} G(s) &= 4 \left[ \frac{s+2}{(s+2)^2+16} \right] \\ &= \frac{4s+8}{s^2+4s+4+16} \\ &= \frac{4s+8}{s^2+4s+20} \end{aligned}$$

Thus, the Laplace transform of  $g(t)$  is,  $G(s) = \frac{4s+8}{s^2+4s+20}$ .